



UNIVERSITY MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE & MATHEMATICS

SOFTWARE ARCHITECTURE
CSE3433

INDIVIDUAL LOGBOOK

GROUP NO. 11

SERVICE-ORIENTED SMART CAMPUS SUPPORT SYSTEM

Prepared by:
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Prepared for:

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[BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)
WITH HONOURS]
SEMESTER 2 2025/2026

1.0 Team Information

Name	Matric No	Role
SARAH AMIRA HERI	S75812	Project Manager & System Analyst
ELSA FADLIN BINTI MERSING	S75412	Frontend Developer & Tester
SITI AISYAH ARWIAH BINTI YUNDI	S74256	Backend Developer & Database Designer

2.0 Weekly Progress Report

Week 4

Item	Description
Week	Week 4 (2-8 March2026)
Weekly Activities	Brainstorm the services used in Smart Campus System
Individual Contributions	Assign task to groupmates, collected the common requirements for the system
Challenges	Understanding the theory of Service Oriented Architecture
Action Plans	Researched on the topic Service Oriented Architecture to gain better understanding

Week 5

Item	Description
Week	Week 5 (9 - 15 March 2026)
Weekly Activities	Drafted the first deliverable project proposal report for submission on 16th April 2026.
Individual Contributions	Finished the Service or Component Identification part of proposal report
Challenges	A bit loss and couldn't find the proper component that fit for the project
Action Plans	Group review with lecturer on the proposal, asked assistance from peers, and searching proper component online or through prompting AI

Week 6

Item	Description
Week	Week 6 (16 - 22 March 2026)
Weekly Activities	Finalized the project proposal report
Individual Contributions	Making System Context Diagram and its explanation and help double checking the report
Challenges	Overcomplicating the the module for the system context diagram
Action Plans	Refer to groupmates opinion for simplifying

Week 7

Item	Description
Week	Week 7 (23 - 29 March 2026)
Weekly Activities	Lead the deign the early SOA architecture for the system
Individual Contributions	Overview the development of the high-level architecture diagram
Challenges	Difficulty in establishing clear communication between services
Action Plans	Research service integration technique

Week 8

Item	Description
Week	Week 8 (30 March - 5 April 2026)
Weekly Activities	Evaluated architectural alternatives and finalized SOA adoption
Individual Contributions	Refining the service interaction diagram
Challenges	Implementing SOA architecture in the system
Action Plans	Referred to YouTube for tutorial

Week 9

Item	Description
Week	Week 9 (6 - 12 April 2026)
Weekly Activities	Overview the development of UML diagrams
Individual Contributions	Checked the System Context Diagram, Component Diagram and Interaction Diagram
Challenges	Inconsistency in between diagrams
Action Plans	Double checking and editing the UML diagrams with team members

Week 10

Item	Description
Week	Week 10 (13 - 19 April 2026)
Weekly Activities	Designed database structure and service integration workflow
Individual Contributions	
Challenges	
Action Plans	

Week 11

Item	Description
Week	Week 11 (20 - 26 April)
Weekly Activities	Overview the prototype implementation plan
Individual Contributions	
Challenges	
Action Plans	Begin developing modules according to SOA design

Week 12

Item	Description
Week	Week 12 (27 April - 3 May 2026)
Weekly Activities	Implemented system prototype and service components
Individual Contributions	Helps establishing databases with the system
Challenges	Different password for the group member local database, making testing difficult
Action Plans	Separate the connection into its own file

Week 13

Item	Description
Week	Week 13 (4 - 10 May 2026)
Weekly Activities	Finished Project and checked all assets before the submission report.
Individual Contributions	Cleaning the code of the project and fixes visible bugs before submission
Challenges	Version code clashing, making new bugs
Action Plans	Ensure new codes are pushed inside repository properly

3.0 GenAI Usage Record

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
3	Perplexity	Understand the software architecture patterns	“List out and explain simply Software Architecture Patterns”	An explanation of software architecture patterns in layman terms	Summarizing the types of software architecture patterns
4	Deepseek	Requirement gathering	“Functional Requirement of a Smart Campus System”	A list of the common requirement of the system with brief explanation	Requirement gathering of the project
		Requirement gathering	“Non-functional requirement of a Smart Campus System”	A list of the common non-functional of the system with simple explanation	Requirement gathering of the project
	Gemini	Component Identification	“What’s the common component used in SOA”	Component usually used in SOA	Research on SOA
		Integrating component Identification (1)	“What service is usually use for user module for a system in microservices?”	Listed common services used including keycloak etc	Research on project

		Integrating component Identification (2)	“What service is usually use for booking module for a system in microservice s?”	Listed common services used	Research on project
		Integrating component Identification (3)	“What database for system in microservice s?”	Listed common services used including Kafka and rabbit etc	Research on project